■ WARUNG MADURA DIGITAL

MASTER DOCUMENTATION: PANDUAN PENGGUNA & SPESIFIKASI TEKNIS LENGKAP

Unified Bilingual Manual (Indonesia - English) with Real UI Screenshots & Deep Technical Specs

---

■ CATATAN LANDASAN WEBINAR DEMO PYTHON | PYTHON DEMO WEBINAR CONTEXT:
■■ **Bahasa Indonesia:** Aplikasi ini dikembangkan sebagai bentuk pembuktian nyata dari **Webinar Demo Python** dengan paradigma *'1 Aplikasi Bisa Untuk Semua'* (One Unified App). Mengintegrasikan fungsi Kasir POS, Inventaris Stok, Pencatatan Kehabisan Barang, dan Laporan Analisis ke dalam satu platform Web Streamlit untuk studi kasus **Warung Madura 24 Jam**.
■■ **English:** Built as a practical application of the **Python Demo Webinar**, proving the *'One Application Fits All'* paradigm. It unifies POS Cashier, Inventory Management, Out-of-Stock Demand Logging, and Business Intelligence into a single Streamlit platform tailored for a **24/7 Warung Madura** store.

Nama Aplikasi / App Name	Warung Madura Digital (Catatan Transaksi & Stok)
Technology Stack	Python 3.10+ | Streamlit | Pandas | SQLite3 | Plotly Express
Cakupan Dokumentasi	Master Combined (User Manual + Full Technical Specs) Bilingual
Tanggal / Date	Agustus / August 2026
Versi Dokumen	3.0.0 Master Final Edition (With Code Structure & DAL API Specs)

DAFTAR ISI / TABLE OF CONTENTS:

CH. 1: ARSITEKTUR PERANGKAT LUNAK & STRUKTUR KODE (SOFTWARE ARCHITECTURE)

■■ **Bahasa Indonesia:** Aplikasi `Warung Madura Digital` mengadopsi pola arsitektur **Two-Tier Layered Architecture** yang memisahkan secara tegas antarmuka pengguna (*Presentation Layer*) di file `app.py` dan lapisan akses data (*Data Access Layer - DAL*) di file `database.py`.

■■ ***English:*** *The `Warung Madura Digital` application adopts a **Two-Tier Layered Architecture** strictly separating the user interface (*Presentation Layer*) in `app.py` and the Data Access Layer (*DAL*) in `database.py`.*

File / Module	Peran & Komponen Utama / Primary Role	Deskripsi Teknis / Technical Description
app.py	Presentation Layer & UI Controller	Mengelola layout Streamlit, CSS styling, session state cart, routing menu sidebar, & visualisasi Plotly charts.
database.py	Data Access Layer (DAL) & SQL Helper	Mengelola koneksi SQLite `warung_madura.db`, inisialisasi 5 tabel, transaksi atomic, & query agregasi pandas.
warung_madura.db	SQLite Embedded Database File	Penyimpanan file terenkapsulasi zero-server yang menyimpan 5 tabel relasional toko.

CH. 2: DASHBOARD UTAMA & OPERATIONAL METRICS

■■ **Bahasa Indonesia:** Layar utama menampilkan 4 KPI Card (Omset Total/Harian, Total Jenis Produk, Peringatan Stok Menipis/Habis, dan Log Barang Dicari) serta 2 Grafik Plotly cepat.
■■ **English:** The main dashboard presents 4 KPI Cards (Total/Today Revenue, Product Count, Stock Warning Alerts, and OOS Log Count) plus 2 Plotly quick charts.

Gambar / Figure: Tampilan Dashboard Utama Warung Madura Digital | *Main Dashboard UI View*

■■ IMPLEMENTASI TEKNIS / TECHNICAL IMPLEMENTATION:

■■ **Bahasa Indonesia:** `app.py` memanggil `db.get_all_transaksi_df()`, `db.get_barang_df()`, & `db.get_log_permintaan_df()`. Metrik harian dihitung via Pandas `df_transaksi[tanggal_only == today_str]['total_harga'].sum()`. Grafik dibuat dengan `plotly.express.pie` & `bar`.

■■ ***English:*** *`app.py` queries `db.get_all_transaksi_df()`, `db.get_barang_df()`, & `db.get_log_permintaan_df()`. Daily metrics are computed via Pandas date filtering. Charts are dynamically generated using `plotly.express.pie` & `bar`.*

CH. 3: MODUL KASIR & WORKFLOW TRANSAKSI (POS CASHIER & ACID CODE)

■■ **Bahasa Indonesia:** Pencarian produk instan, keranjang belanja multi-item, kalkulator kembalian otomatis, dan fitur ekspander *Fast Out-of-Stock Logger* di layar kasir.
■■ **English:** Instant product search, multi-item cart, automatic change calculator, and an embedded *Fast Out-of-Stock Logger* form.

Gambar / Figure: Layar Kasir POS & Form Fast Out-of-Stock Logger | *POS Cashier Interface & Fast OOS Logger Form*

■■ TRANSAKSI ATOMIC & POTONGAN KODE (`database.py`):

■■ **Bahasa Indonesia:** Menggunakan transaksi atomic ACID (`commit` / `rollback`) saat memproses keranjang belanja.

```
def process_transaction(items_cart, metode_pembayaran="Tunai", catatan=""):
    conn = get_connection()
    cursor = conn.cursor()
    try:
        # 1. Insert Header Transaksi (transaksi)
        cursor.execute("INSERT INTO transaksi VALUES (?, ?, ?, ?, ?)", (kode_tx, now, total, metode, cata
tan))
        id_tx = cursor.lastrowid
        # 2. Insert Detail Item & Update Stok Physical (detail_transaksi & barang)
        for item in items_cart:
            cursor.execute("INSERT INTO detail_transaksi VALUES (...)", (id_tx, item['id'], ...))
            cursor.execute("UPDATE barang SET stok = stok - ? WHERE id = ?", (item['qty'], item['id']))
        conn.commit()  # Simpan permanen jika seluruh item berhasil
        return True, "Transaksi berhasil dicatat!"
    except Exception as e:
        conn.rollback() # Batalkan seluruh perubahan jika terjadi error!
        return False, str(e)
    finally:
        conn.close()
```

CH. 4: DATA BARANG & KATEGORI MASTER (INVENTORY & MARGIN MATH)

■■ **Bahasa Indonesia:** Pengelolaan master barang & kategori. Hitungan margin profit (Harga Jual - Harga Beli) serta Badge Status Stok (■ HABIS, ■ MENIPIS ≤5, ■ AMAN) dihitung otomatis.
■■ **English:** Product & category master CRUD. Profit margins (Selling - Cost Price) and Stock Badges (■ OOS, ■ LOW ≤5, ■ SAFE) are auto-calculated.

Gambar / Figure: Kelola Inventaris Produk & Master Kategori | *Inventory Management & Master Category Tab*

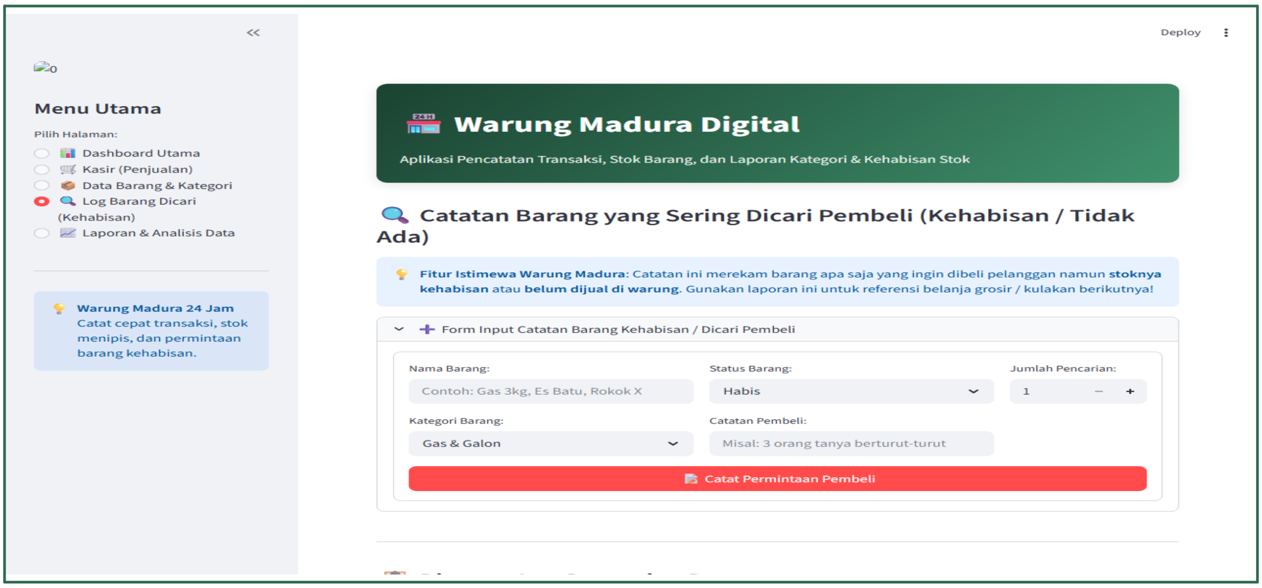
■■ FUNGSI DAL UNTUK INVENTARIS / DAL FUNCTIONS:

■■ **Bahasa Indonesia:** `add_barang()`, `update_barang()`, `delete_barang()`, `add_kategori()`, `delete_kategori()`. Penghapusan kategori dilindungi query `SELECT COUNT(*)` untuk memastikan tidak ada barang yang kehilangan referensi FK.

■■ ***English:*** *`add_barang()`, `update_barang()`, `delete_barang()`, `add_kategori()`, `delete_kategori()`. Category deletion is protected by a `SELECT COUNT(*)` check to ensure no orphaned FK references.*

CH. 5: LOG BARANG DICARI (OUT-OF-STOCK DEMAND ANALYTICS)

■■ **Bahasa Indonesia:** Merekam barang yang dicari pelanggan tapi stoknya habis atau belum dijual. Tabel Ranking mengurutkan pencarian terbanyak sebagai acuan belanja grosir / kulakan.
■■ **English:** Records items requested by customers that were out-of-stock or not offered. The Demand Ranking table guides restocking decisions.



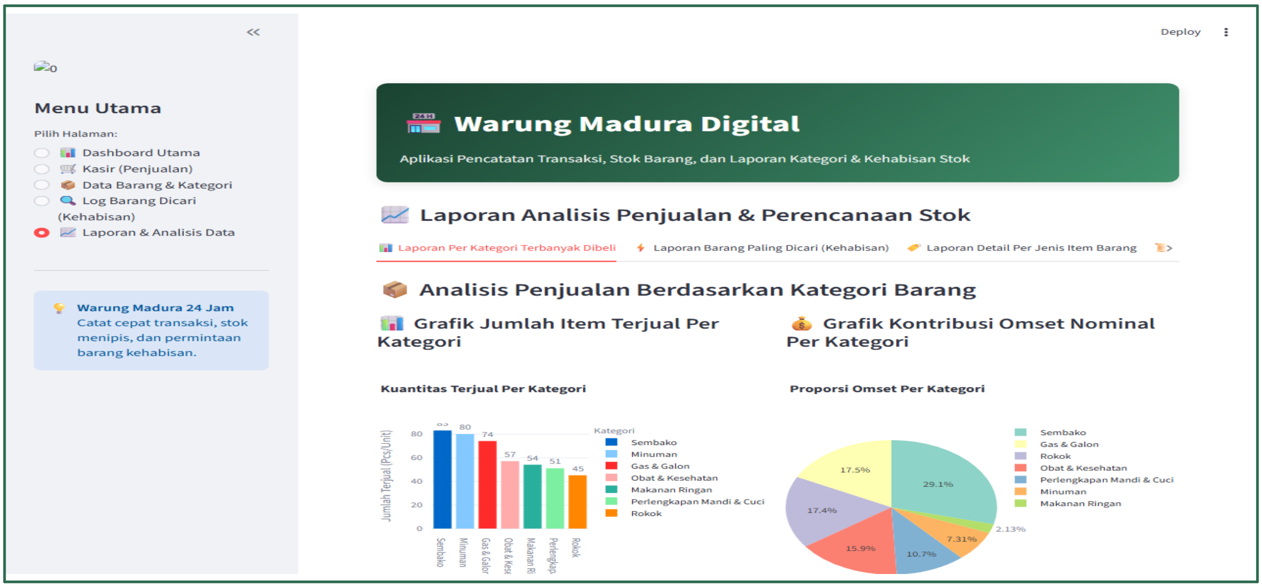
Gambar / Figure: Form Log Pencarian & Ranking Barang Kehabisan Stok | *Search Log Form & Out-of-Stock Demand Ranking*

■■ AGREGASI SQL REKAP DICARI / SQL QUERY:

■■ **Bahasa Indonesia:** Query DAL `get_report_barang_dicari_df()`:
SELECT l.nama_barang, k.nama_kategori, l.status, SUM(l.jumlah_permintaan) AS frekuensi_dicari, COUNT(l.id) AS jumlah_kejadian FROM log_permintaan l LEFT JOIN kategori k ON l.id_kategori = k.id GROUP BY l.nama_barang, k.nama_kategori, l.status ORDER BY frekuensi_dicari DESC

CH. 6: LAPORAN BISNIS & EKSPOR DATA (BUSINESS REPORTS & EXPORT)

■■ **Bahasa Indonesia:** 4 sub-tab laporan interaktif: Analisis Kategori, Barang Kehabisan, Top 10 Terlaris & Profit, serta Riwayat Transaksi & Tombol Unduh CSV.
■■ **English:** 4 interactive report tabs: Category Sales, Out-of-Stock Analytics, Top 10 Best Sellers & Profit, and History with CSV Downloads.



Gambar / Figure: Analisis Grafik Plotly & Laporan Penjualan/Profit | *Plotly Analytics Charts & Sales/Profit Reports View*

■■ AGREGASI PROFIT & EKSPOR CSV / PROFIT MATH & CSV EXPORT:

■■ **Bahasa Indonesia:** Formula profit per item: `SUM(dt.jumlah * (dt.harga_satuan - dt.harga_beli_satuan))`. Ekspor CSV menggunakan `df.to_csv(index=False).encode('utf-8')` pada Streamlit `st.download_button`.

■■ ***English:*** *Item profit formula: `SUM(dt.jumlah * (dt.harga_satuan - dt.harga_beli_satuan))`. CSV export relies on `df.to_csv(index=False).encode('utf-8')` wired to Streamlit `st.download_button`.*

CH. 7: SPESIFIKASI LENGKAP 5 TABEL DATABASE SQLITE (`warung_madura.db`)

■■ **Bahasa Indonesia:** Detail skema struktur kolom-per-kolom untuk seluruh 5 tabel relasional pada `warung_madura.db`.
■■ **English:** Column-by-column schema specification for all 5 relational tables in `warung_madura.db`.

1. Tabel kategori (Master Kategori Barang / Category Master)

Column Name	Data Type	Constraints	Description (ID / EN)
id	INTEGER	PRIMARY KEY AUTOINCREMENT	ID unik kategori / Unique Category ID
nama_kategori	TEXT	UNIQUE NOT NULL	Nama kategori / Category name
deskripsi	TEXT	NULLABLE	Deskripsi kategori / Description

2. Tabel barang (Master Inventaris Produk / Product Inventory Master)

Column Name	Data Type	Constraints	Description (ID / EN)
id	INTEGER	PRIMARY KEY AUTOINCREMENT	ID unik barang / Unique product ID
kode_barang	TEXT	UNIQUE NOT NULL	Kode SKU/Barcode / SKU Code
nama_barang	TEXT	NOT NULL	Nama produk / Product name
id_kategori	INTEGER	FK REFERENCES kategori(id)	Relasi kategori / FK to category
harga_beli	REAL	DEFAULT 0	Harga modal / Cost price
harga_jual	REAL	DEFAULT 0	Harga jual / Selling price
stok	INTEGER	DEFAULT 0	Jumlah stok / Stock quantity
satuan	TEXT	DEFAULT 'pcs'	Satuan unit / Unit (pcs, kg, etc)

3. Tabel transaksi (Header Transaksi Penjualan / Sales Header)

Column Name	Data Type	Constraints	Description (ID / EN)
id	INTEGER	PRIMARY KEY AUTOINCREMENT	ID transaksi / Transaction ID
kode_transaksi	TEXT	UNIQUE NOT NULL	Kode TRX-YYYYMMDD-XXXX
tanggal_transaksi	DATETIME	NOT NULL	Waktu transaksi / Timestamp
total_harga	REAL	NOT NULL	Total belanja / Total price
metode_pembayaran	TEXT	DEFAULT 'Tunai'	Tunai/QRIS/Transfer
catatan	TEXT	NULLABLE	Catatan kasir / Cashier notes

4. Tabel detail_transaksi (Item Rincian Penjualan / Sales Line Items)

Column Name	Data Type	Constraints	Description (ID / EN)
id	INTEGER	PRIMARY KEY AUTOINCREMENT	ID rincian item / Line item ID
id_transaksi	INTEGER	FK REFERENCES transaksi(id)	FK ke header / FK to header
id_barang	INTEGER	FK REFERENCES barang(id)	FK ke produk / FK to product
nama_barang	TEXT	NOT NULL	Snapshot nama / Product name
kategori	TEXT	NOT NULL	Snapshot kategori / Category
jumlah	INTEGER	NOT NULL	Kuantitas (Qty) / Quantity
harga_satuan	REAL	NOT NULL	Snapshot harga jual / Unit price
harga_beli_satuan	REAL	DEFAULT 0	Snapshot harga modal / Cost price
subtotal	REAL	NOT NULL	Subtotal (Qty * Harga Satuan)

5. Tabel log_permintaan (Pencatatan Kehabisan Barang / Out-of-Stock Demand Log)

Column Name	Data Type	Constraints	Description (ID / EN)
id	INTEGER	PRIMARY KEY AUTOINCREMENT	ID log / Log ID
tanggal	DATETIME	NOT NULL	Waktu stempel / Timestamp
nama_barang	TEXT	NOT NULL	Nama barang dicari / Item name
id_kategori	INTEGER	FK REFERENCES kategori(id)	FK ke kategori / FK to category
jumlah_permintaan	INTEGER	DEFAULT 1	Qty ditanyakan / Requested Qty
status	TEXT	DEFAULT 'Habis'	'Habis' / 'Belum Dijual'
catatan	TEXT	NULLABLE	Catatan konteks / Notes

CH. 8: REFERENSI LENGKAP FUNGSI API DATA ACCESS LAYER (`database.py`)

■■ **Bahasa Indonesia:** Tabel lengkap referensi 13+ fungsi Data Access Layer (DAL) pada modul `database.py`.
■■ **English:** Complete reference table of all 13+ Data Access Layer (DAL) functions in `database.py`.

Function Name	Parameters	Return Type	Description (ID / EN)
init_db()	None	None	Inisialisasi 5 tabel SQLite & seeder otomatis jika DB kosong. *Initializes 5 tables & seeds sample data if DB is empty.*
get_connection()	None	sqlite3.Connection	Membuka koneksi ke `warung_madura.db` dengan row_factory SQLite.Row. *Opens SQLite connection with row_factory.*
seed_sample_data()	conn	None	Seeding data awal kategori, 17 barang, transaksi 14 hari, & log permintaan. *Seeds initial categories, products, sales history, & demand logs.*
get_kategori_df()	None	pandas.DataFrame	Query seluruh master kategori diurutkan A-Z. *Fetches all master categories sorted alphabetically.*
get_barang_df()	None	pandas.DataFrame	Query JOIN master barang & kategori beserta margin profit (jual - beli). *Fetches product list joined with categories & margin math.*
add_kategori()	nama, deskripsi	(bool, str)	Menambah kategori baru dengan penanganan UNIQUE constraint. *Inserts new category with UNIQUE constraint handling.*
delete_kategori()	kat_id	(bool, str)	Hapus kategori dengan proteksi jika masih dipakai di barang. *Deletes category with FK usage validation check.*
add_barang()	kode, nama, id_kat, h_beli, h_jual, stok, satuan	(bool, str)	Tambah barang baru dengan validasi kode barang unik. *Inserts new product with unique SKU code check.*
update_barang()	id_brg, nama, id_kat, h_beli, h_jual, stok, satuan	(bool, str)	Memperbarui data barang & stok di database. *Updates product info & stock levels in SQLite.*
delete_barang()	id_barang	(bool, str)	Menghapus data barang dari tabel barang. *Deletes product entry from barang table.*
process_transaction()	items_cart, metode, catatan	(bool, str)	Eksekusi transaksi atomic ACID: insert header, insert detail, & update stok. *Executes atomic transaction: insert header, details, & update stock.*
log_barang_dicari()	nama, id_kat, qty, status, catatan	(bool, str)	Menyimpan catatan barang yang dicari pelanggan ke log_permintaan. *Logs customer requested missing items to log_permintaan.*
get_report_kategori_df()	None	pandas.DataFrame	Agregasi SQL SUM(qty), SUM(omset), & SUM(profit) per kategori. *SQL aggregation of qty, revenue, & profit per category.*
get_report_item_df()	None	pandas.DataFrame	Agregasi SQL penjualan & profit margin per jenis item barang. *SQL aggregation of sales & profit margins per item.*
get_report_barang_dicari_df()	None	pandas.DataFrame	Query ranking barang paling dicari (kehabisan stok) dikelompokkan A-Z. *Queries ranking of most requested missing items.*
get_all_transaksi_df()	None	pandas.DataFrame	Query seluruh riwayat transaksi penjualan diurutkan waktu terbaru. *Queries full sales transaction history sorted descending.*

CH. 9: INSTALASI, DEPLOYMENT & MAINTENANCE BACKUP STRATEGY

1. Prerequisites & Installation Commands:

```
pip install -r requirements.txt
python -m streamlit run app.py
```

2. Strategi Backup Database / Database Maintenance Strategy:

■■ **Bahasa Indonesia:** Karena SQLite bersifat *serverless* & *zero-configuration*, seluruh data bisnis tersimpan dalam file warung_madura.db. Lakukan copy/backup file ini ke Google Drive / Flashdisk secara berkala.
■■ **English:** Because SQLite is *serverless* & *zero-configuration*, all business data resides in warung_madura.db. Regularly copy/backup this file to external cloud storage or USB drives.

MASTER DOCUMENTATION — WARUNG MADURA DIGITAL (COMPLETE EDITION)

Hasil Praktik Webinar Demo Python | Practical Result of Python Demo Webinar